

MARK CHEN

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*Alpha-numerical & (reverse-)chronological orders are applied whenever/however appropriate; nothing is ordered by any form of priority

EDUCATION (DOCTORATE AND VISITING AFFILIATIONS DURING PH.D.)

STANFORD UNIVERSITY – SCHOOL OF ENGINEERING

STANFORD, CA

Ph.D. in Computer Science (Minor in Electrical Engineering); M.S. in Passing

Expected: SEPT 2026 – JUNE 2031

- **Advisors:** Prof. Emma Dauterman; **Committee:** Profs. Dan Boneh, Ari Juels (Cornell Tech), David Mazières
- **Focus:** Theory and Practical Aspects of Cryptography for Better ML Models and Systems (Checkability, Accountability, etc.)

CORNELL TECH

NEW YORK, NY

Visiting PhD Student + Affiliate of IC3

SUMMER 2026, PART-TIME/REMOTE: SEPT 2026 – PRESENT

- **Advisor:** Prof. Ari Juels; **Focus:** Applying Blockchain and Hardware Security Techniques to Safeguard Agentic AI

EDUCATION (EARLIER)

COLUMBIA UNIVERSITY – FU FOUNDATIONS SCHOOL OF ENGINEERING

NEW YORK, NY

B.S. in CS + Math (Major GPA: 4.04, cGPA: 4.00, Magna Cum Laude + Tau Beta Pi: Top 12.5%) MAY 2021 – DEC 2024; JAN – MAY 2026

- **Advisors:** Profs. Xi Chen, Roxana Geambasu, Tal Malkin, and Toniann Pitassi; **Focus:** Theory and Systems for Applied Cryptography
- **Highlights:** Contributed to works that were published at SOSP’26 and STOC’26 (winning the Best Paper Award)

PUBLICATIONS

1. **Mark Chen**, Xi Chen, Hao Cui, William Pires, Jonah Stockwell. “Boolean Function Monotonicity Testing Requires (Almost) $n^{1/2}$ Queries”. **Best Paper Award** at ACM Symposium on Theory of Computing (STOC’26). [arXiv]
2. Pierre Tholoniati, Alison Caulfield†, Giorgio Cavicchioli†, **Mark Chen**†, Benjamin Case, Asaf Cidon, Roxana Geambasu, Mathias Lécuyer, Martin Thomson. “Big Bird: Resilient Privacy Budgeting Across Untrusted Web Domains”. *Symposium on Operating Systems Principles (SOSP’26)*. [arXiv]

(Authors are alphabetically sorted by default; otherwise; *-marked authors have equal contributors; †-marked authors are non-PhD student authors with considerable contributions sorted alphabetically by Columbia DP Infra Lab convention with equal contributions not implied)

MANUSCRIPTS

1. Giorgio Cavicchioli, **Mark Chen**†, Navid S. Pargoo†, Shuren Xia†, Xiaotian Zhou†, Roxana Geambasu, Jason Nieh, Jorge Ortiz. “CityOS: Privacy Architecture for Urban Sensing”. Submitting to *USENIX Symposium on Operating Systems Design and Implementation (OSDI’27)*. [arXiv]
2. Peihan Liu, Alison Caulfield†, **Mark Chen**†, Rachel Cummings, Roxana Geambasu, Mathias Lecuyer, Pierre Tholoniati. “Adaptive Target-Charging with Privacy Filters and Individual Accounting”. Submitted to *Neural Information Processing Systems (NeurIPS’26)*.

RESEARCH EXPERIENCE (OTHER THAN PRIMARY PhD PROGRESS)

THE INITIATIVE FOR CRYPTOCURRENCIES AND CONTRACTS (IC3, PI: PROF. ARI JUELS)

CORNELL UNIVERSITY (CORNELL TECH)

ML Accountability

MAR 2026 – PRESENT

- Constructed and wrote security proofs for a TEE-based protocol for “certified probe” that certificates a model’s properties with grey-box access (i.e. through looking at its I/Os, TODO), under the hybrid model; mapped over concepts from sealed-glass proofs and runtime enforcement
- Implemented the sealed-glass proof in Rust, with modernized TEE attestations from TDX, TODO

Unstoppable Agents

MAR 2026 – PRESENT

- Proposed the inductively defined game over time-steps of the running of the rogue agent to define the unstoppable game; the game is to detect and shutdown instances of N cloud providers based solely on views adaptive generated from a set rogue goal leveraging what a typical cloud telemetry would allow and the winning condition is when instances of an agent on these cloud providers are completely shutdown

INDEPENDENT PROPERTY TESTING RESEARCH (PI: PROF. XI CHEN)

COLUMBIA UNIVERSITY

Boolean function monotonicity testing (publications section item 1)

FEB – NOV 2025

- Helped formalize the adaptive tester behavior on the multi-level generalization of the original Talagrand construction [CXW’17], and identify independent per-leaf secret sampling as the key obstacle to motivate our multilevel global-secret construction
- Independently proved most intermediate claims and the “good outcome” induction bounding layer-by-layer information leakage in the new construction, contributing to the nearly tight $\Omega(n^{1/2-c})$ adaptive lower bound. The same construction and similar proofs also settled, up to a sub-polynomial gap, the complexity of testing both monotonicity and unateness in the relative-error model
- Took the initiative to pursue and proved the first version of adaptivity hierarchy lower bound for the new construction (i.e. for any constant round of adaptivity, our construction gives tight lower bounds), though it only worked for frontier-level information leakage

DIFFERENTIAL PRIVACY (DP) INFRASTRUCTURE LAB (PI: PROF. ROXANA GEAMBASU)

COLUMBIA UNIVERSITY

Big Bird: effective cross-site differential privacy guarantee for replacing Cookie for ads (publications section item 2)

JAN – DEC 2025

- Independently formulated & proved theories for global privacy budget management for the Sybil-resistant system design, including the current data modeling, sensitivity bounds, resilience-to-DoS-depletion theorem against Sybil attacks, and so on, up to refinements
- Implemented 1/3 of the general DP budget API in **Rust** based on theoretical designs, being used by several parallel projects to (a) ensure privacy, (b) mitigate large-scale Sybil attacks, (c) support optimization queries, and (d) improve utility of privacy budget
- Worked with resilience in short timeframes, as one of the two main technical contributors, to evaluate learning with labeled proportions against different DP architectures for the presentations to the W3C Private Advertising Technology Working Group (PATWG); conducted sprint sessions and identified fixes to several unexpected outputs. Helped convince PATWG of our Big Bird design with these experiment results for potential adoption in W3C's browser ads attribution API standard for a very broad impact
- Developed "quota counts" and user actions for per-user-action caps on site visits, separating benign behaviors from Sybil ones; independently incorporated these per-user-action quotas into Big Bird's privacy and graceful-degradation analyses

Improving target charging techniques (TCT) analyses with individual RDP filters (Submitting to ICML '26)

DEC 2024 – PRESENT

- Independently modeled TCT trials as a negative-binomial process to derive a new moment bound, and initiated a Rényi-DP (RDP) target-charging analysis that improves the dependence on the failure probability, δ , compared to the original pure-DP analysis
- Extended the RDP decomposition to handle query- and budget-adaptive workloads, enabling TCT to support realistic interactive queries and dynamic privacy budgets while maintaining tight privacy guarantees under point-wise composition of RDP filters
- Took initiative to connect individual RDP literature and refined the $\mathcal{T}$ -target RDP filter moment bounds via per-individual granularity

CityOS: privacy API architectures for smart city infras (publications section item 3)

OCT 2024 – APR 2026

- Designed and implemented continual-release counting APIs on top of OpenDP, after a thorough review of matrix-factorization methods, to enable private counting in CityOS's single-locality aggregation API; validated these APIs on parking-availability prediction and differentially private releases of Florida traffic-sensor data
- Identified a previously unaddressed assumption in [Privid](#) where objects can be counted at most once per frame, revealing a new class of association attacks that enable double-counting by untrusted image-detection algorithms that CityOS allows users to install. Currently researching architectural designs to bound the resulting inflation in event-level sensitivity and utility loss in CityOS
- System designed privacy accounting in the heterogenous city environment using odometers via self-identification

CRYPTOGRAPHY LAB (PI: PROF. TAL MALKIN & PROF. TONIANN PITASSI)

COLUMBIA UNIVERSITY

Average-case TFNP separations in the random oracle model (ROM)

OCT 2024 – JUNE 2025, SEPT 2025 – NOV 2025

- Surveyed and presented average-case hardness for TFNP subclasses in the random-oracle model via recursion of long computations made incremental using Fiat–Shamir; observed that all known hardness for bidirectional classes demands computational assumptions, motivating our focus on PPA as the highest such class with bidirectionality
- Studying leveraging the random oracle VDF query complexity impossibility to pinpoint a gap between unconditional constructions and constructions based on computational assumptions; currently exploring pebbling-based approaches to VDF-like constructions that hide global direction at each local evaluation while maintaining parallel-round hardness

POSTERS, TALKS, AND PRESENTATIONS SEE: [HTTPS://MYCERN.GITHUB.IO/TAB/RESEARCH_AND_PROJECTS.HTML](https://mychern.github.io/tab/research_and_projects.html)

- [Big Bird] Evaluating Optimization Queries: Methodology and Preliminary Results [[PATWG GitHub issue](#), [Slides](#)]
- [Individual Sparse Vector via Target Charging] Adaptive Individual Accounting with Target Charging technique [[Poster](#)]
- [Big Bird][CityOS] Privacy-Preserving Architecture: From W3C Standards to Smart Cities [[Poster](#)] (**Unanimous Top-Band Rating**)
- [Average-Case Hardness of TFNP classes in ROM] Proof of Sequential Work / Verifiable Delay Functions in ROM [[Typed](#), [Written](#)]
- [Boolean Function Monotonicity Testing] STOC'26 Virtual Session Poster Talk [Poster]
- [Boolean Function Monotonicity Testing] Hour-Long Talk at the Cornell Theory Group (in Ithaca) [Slides]

TEACHING EXPERIENCE

COLUMBIA UNIVERSITY

NEW YORK, NY

Teaching Assistant for COMS 3262 Intro to Cryptography (Expected Enrollment: 30+ Students)

SPRING 2026

Teaching Assistant for CSOR 4231 Analysis of Algorithms (Final Enrollment: 130+ Students)

SPRING 2026

Instructor for the high school program by Center for Smart Streetscapes (CS3)

SUMMER 2025

- Explained fingerprinting, cookies, etc., and research replacing Cookie w/ a differentially private system; supervised capstone projects

Teaching Assistant for COMS 4236 Intro to Computational Complexity (Final Enrollment: 25 Students)

FALL 2023

- As a team, the teaching staff received high reviews from online student evaluations for helpfulness and approachability

- Held weekly office hours to answer student questions, graded and contributed to making questions for homeworks and exams

Teaching Assistant for COMS 4231 Analysis of Algorithms (Final Enrollment: 200+ Students)

FALL 2023

- Held weekly office hours to answer student questions, graded homeworks and exams, organized logistics and proctored exams

CLASS PROJECTS SEE: [HTTPS://MYCERN.GITHUB.IO/TAB/RESEARCH_AND_PROJECTS.HTML](https://mychern.github.io/tab/research_and_projects.html)

- "Equivalences between TFNP^{dt} and Low-Level Proof Systems with an example in $\text{PPA}^{\text{dt}} \cong F_2\text{-Nullstellensatz}$ ", Spring 2025 [[PDF](#)]
- "Fractional Pseudorandom Generators and Bounded Fourier Tails: A Literature Review", Spring 2024 [[PDF](#)]
- "Bounding Threshold Formulas using Multiparty Communication Complexity Techniques", Spring 2024 [[PDF](#)]
- "Path to Secure Reinforcement Learning - Explicit Construction as an **IP** and Cryptographic Implications", Spring 2024 [[PDF](#)]

INDUSTRY EXPERIENCE

GOOGLE ADS

NEW YORK, NY

SWE Intern on Ads Privacy and Data Governance — Privacy Design Docs (PDDs) Review System (12 WEEKS) JULY – OCT 2025

- Built **Java data pipeline** for parsing 2700+ PDDs and developed **Java RPC services** and **CLI tools** for using Gemini APIs to ingest legacy PDDs and suggest answers to the enhanced PDDs (EPDDs); shortened each PDD-to-EPDD migration from 1+ week to 6 hours

YOUTUBE

VARIED LOCATIONS, 3X INTERNSHIPS

SWE Intern on Nitrate for Pay-Gated Content Platform (PGP)

MOUNTAIN VIEW, CA; (14 WEEKS) MAY – AUG 2024

- C++ data pipeline for YouTube PGP recommendations on Home to enable transactional features, experimented to be some of the most important features by Pearson correlations with click per impression; iterated several **Python** multi-task models

SWE Intern on Over-the-Top for Digital Video Recording (DVR)

SAN BRUNO, CA; (14 WEEKS) MAY – AUG 2023

- Developed auto, async DVR playability testing system (C++ & **Python** to schedule, test and alert; **Spanner** to coordinate) on iOS, Android, Web & Home devices to reduce mean time to detect DVR processing outages from days to hours

STEP Intern on Commerce for Non-Recurring Plans (NRP)

MOUNTAIN VIEW, CA; (12 WEEKS) JUNE – SEPT 2022

- C++ RPC processor to list offers, verify purchases, and update entities for NRP; enabled \$50 millions revenue

K2VC (\$2BN ANGEL); **ZHENFUND** (ANGEL FUND OF SEQUOIA CAPITAL) - *VC Analyst*

BEIJING, CHINA; SEPT – DEC 2020; JAN – JUNE 2021

OTHER EXPERIENCE (REMOTE / PART-TIME ETC.)

QUANTITATIVE SWE RESEARCHER (**MATCHBOXDAO**, Brownie game dynamics analysis)

(REMOTE, TEAM IN ISRAEL, 4 WEEKS) JAN 2023

WINTER SWE INTERN (**MICROSOFT**, C# Azure Navigation Chatbot)

(REMOTE, TEAM IN SHANGHAI, 4 WEEKS) JAN 2022

RECOGNITIONS

- Best Paper Award at STOC'26
- Initiative for Cryptocurrencies and Contracts (IC3) Hackathon 2nd place winner in 2026 (with a .5 ETH prize) 2nd place in IC3 hackathon (with a .5ETH prize): for the "Trustless Agent" project based on <https://arxiv.org/abs/2606.03771> for agent property auditing; also won the 3rd place for "funniest presentation"
- Stanford School of Engineering (SoE) Fellowship (Free Rotations)
- Cornell Fellowship (Exempted from Teaching Requirements)
- UW CSERF (+\$3,000/year)
- Yale Sterling Prize Fellowship (+\$3,000/year)
- *Named College Scholarships*: UPenn Ben Franklin, JHU Hodson
- *Undergrad*: Putnam Math Score of 21 (Ranked 769/5000+), Google MLthon finalist
- *High School*: Regeneron STS Scholar, ISEF State 2nd Prize, Yau Science Awards CS Global Honorable Mention (Top 10/200+)